ASEN 5090 Lecture 14
Improved GNSS Solutions

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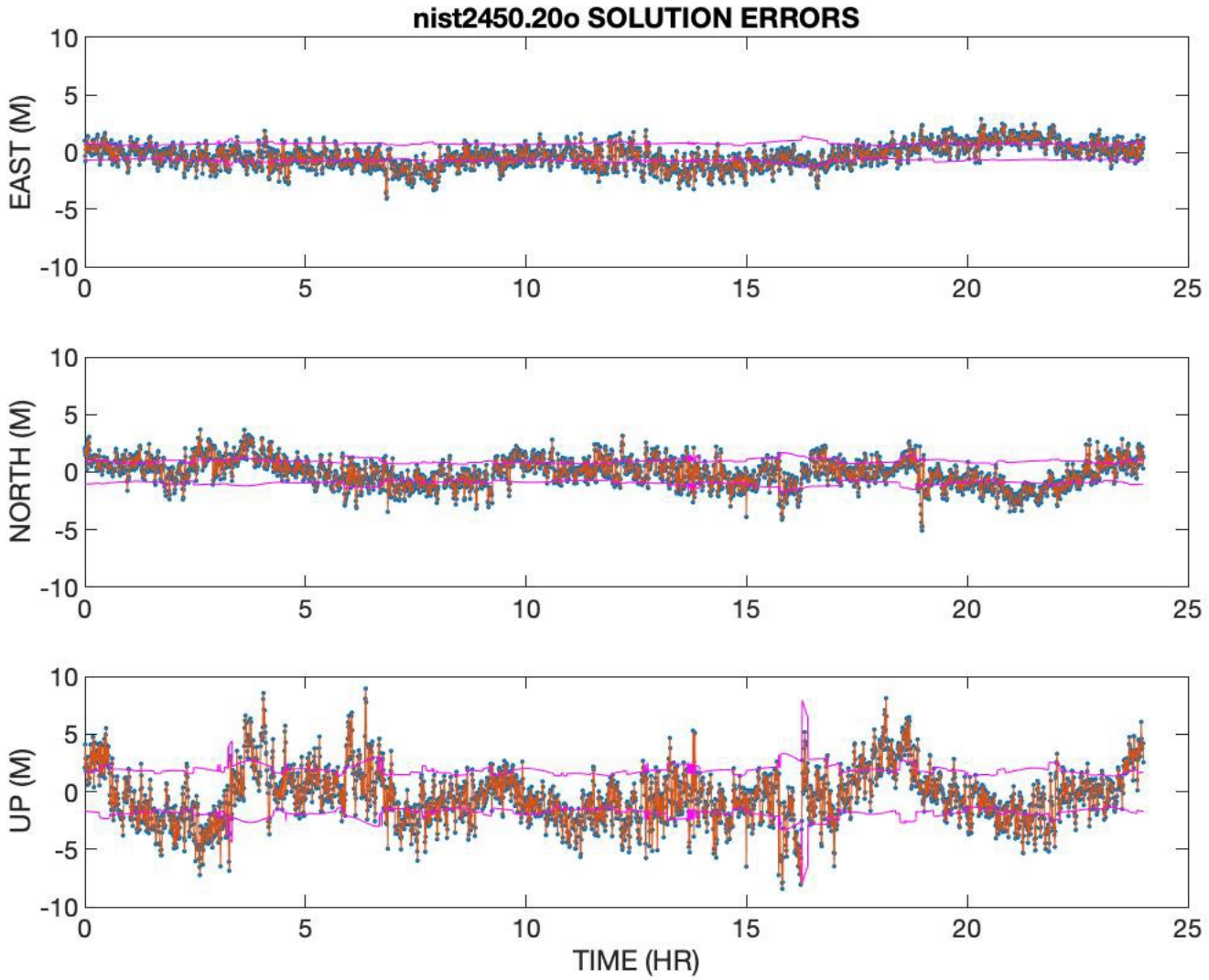
Course schedule

Week	Date	Topic	Reading	Assignments - Due 11pm*
7	10/5/26	L12. Least Squares Solutions	ME Ch 6.A	
	10/7/26	L13. GPS Point Solutions	ME Ch 6.1	
	10/9/26			
8	10/12/26	L14. Improved GPS Solutions		
	10/14/26	L15. Midterm Review & HW Discussion		
	10/16/26			HW5 Due, HW6 Assigned
9	10/19/26	L16. Precise Positioning Methods	ME Ch 7	
	10/21/26	Midterm Exam (no lecture)		Midterm Release 10/21 8AM
	10/23/26			Midterm Close 10/25 11PM
10	10/26/26	L17. Introduction to Signals & Spectra	ME Ch 8	
	10/28/26	L18. GNSS Spectrum	ME Ch 9	
	10/30/26			HW6 Due, HW7 Assigned

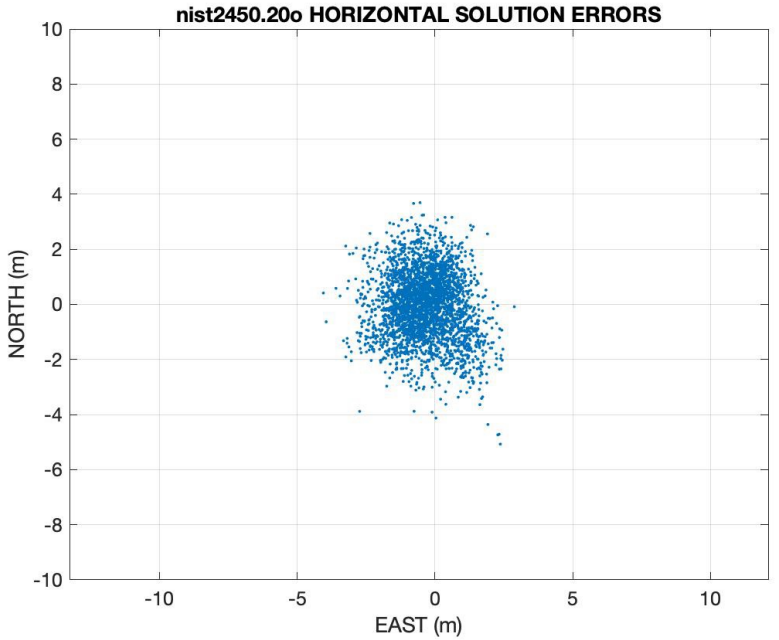
GNSS Performance Metrics

- **Accuracy**: Fundamental measure of a positioning system.
  - Current accuracy:
    - Low-end receivers ~within 10 m
    - WAAS enabled receivers~ within 2m
    - DGPS ~ dm, cm, mm
  - More signals, more frequency, more satellites -> better accuracy
- **Robustness**: Service availability and continuity.
  - Concerns: radio frequency interference, ionosphere disturbances
  - Frequency diversity, higher signal power, higher processing gain -> more robust receivers
- **Integrity**: Assurance that a system is operating correctly and there will be a timely warning if there are signal anomalies
  - Control segment
  - Receiver autonomous integrity monitoring (RAIM) at the user's end

How accurate are the point solutions? NIST Sept 1, 2020 results:

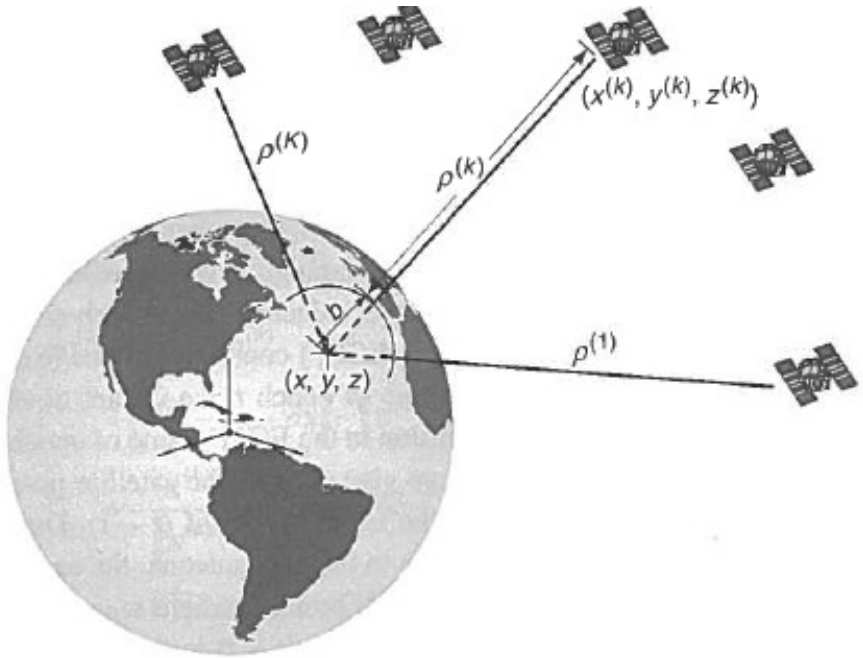


RMS Errors: E 1.1 N 1.2 U 2.4 (m)



We used a poor tropo model and noisy dual frequency corrections

What else can be done to improve GNSS-based positioning?

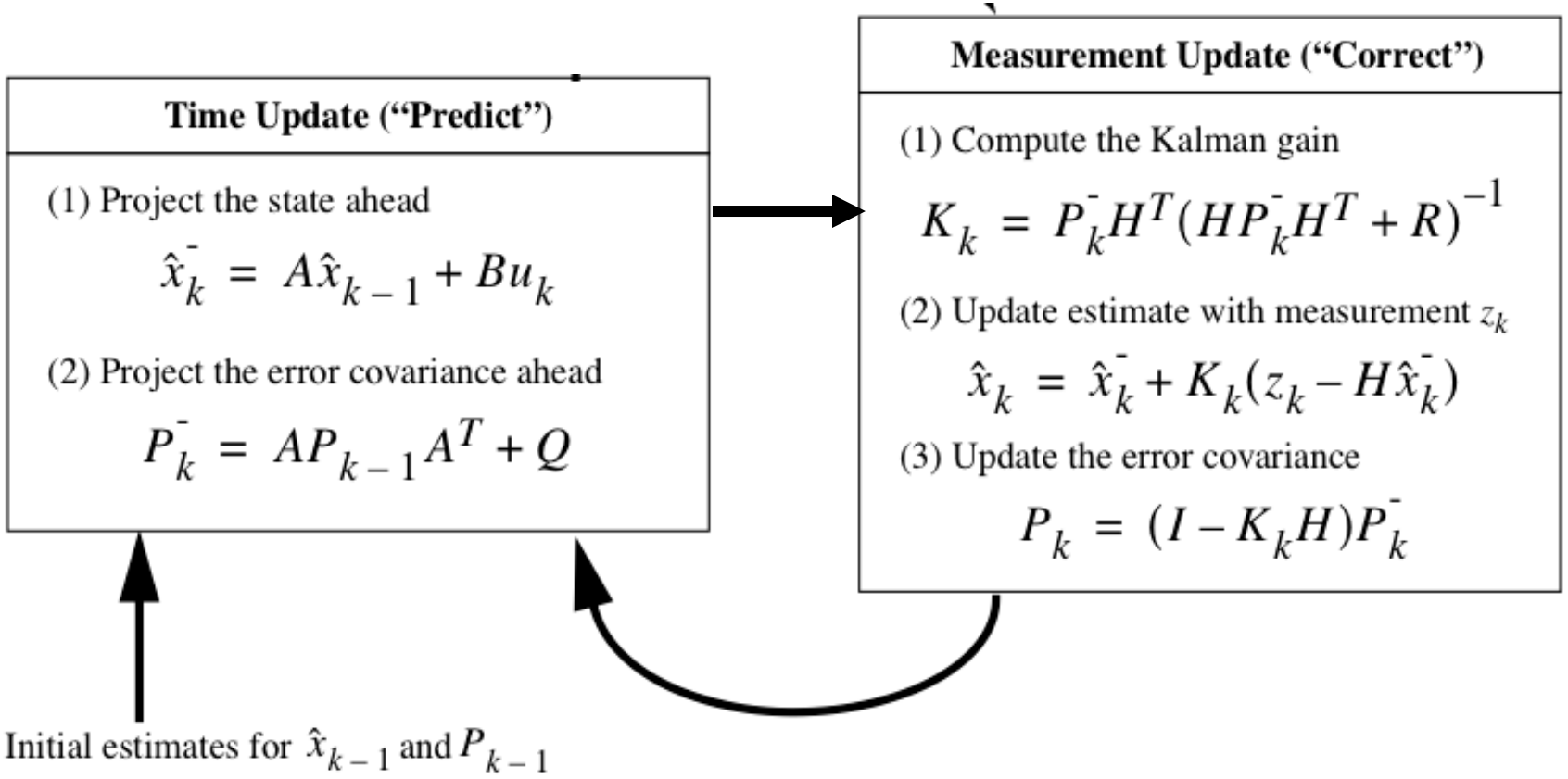


From Misra & Enge

- Multi-GNSS
- Receiver autonomous integrity monitoring (RAIM)
- Kalman Filtering
- Integration with inertial sensors
- Integration with other sensors
- Urban – shadow matching and ray tracing
- Direct position estimation (DPE)
- Other signals of opportunity
- Precise, carrier-based positioning (next week)

Kalman Filter

- Incorporate known dynamics to tie together observations over time
- Optimally weight/balance uncertainty in observations and propagation



Extended Kalman Filter (EKF)

- Nonlinear measurement model
  - Calculate prefit residual using nonlinear measurement model with current state estimate
  - Calculate measurement sensitivity based on current state estimate – used to compute Kalman gain and updated covariance (P)
- Nonlinear dynamical model
  - Propagate state w/nonlinear model
  - Propagate uncertainty (Covariance Matrix P) with linearized model

Integrated GNSS-INS navigation solutions

- Inertial sensors – accelerometers + gyros provide the dynamic propagation
- GNSS provides observations of the navigation state that allow sensor errors to be estimated

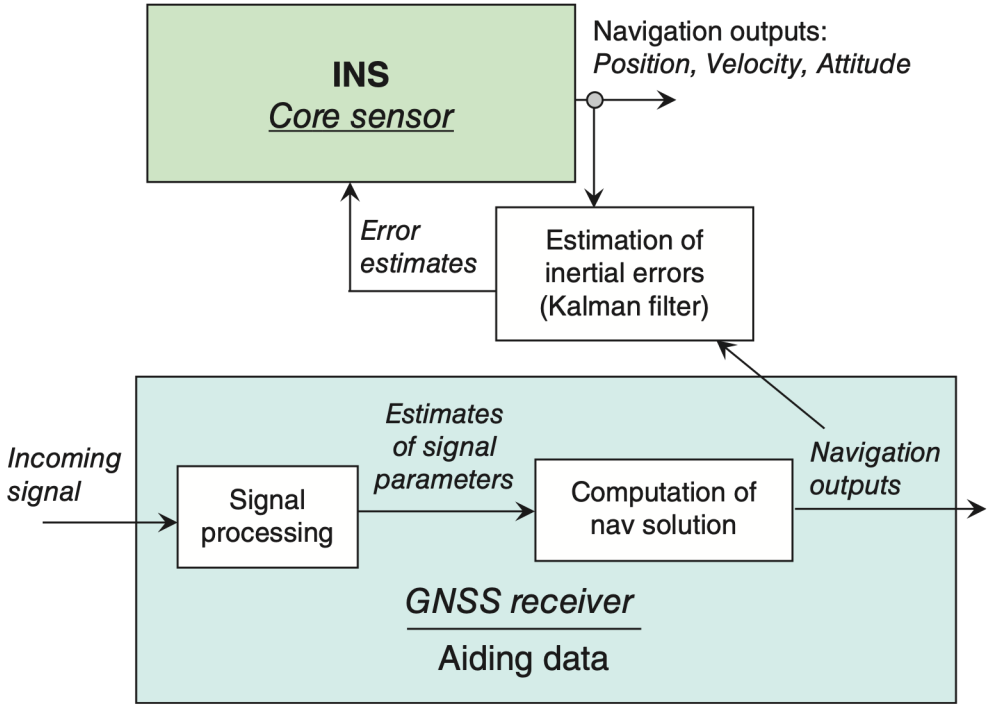


Figure 46.10 Loosely coupled GNSS/INS mechanization. GNSS navigation solution (v
navigation outputs.

Integrated GNSS-INS navigation solutions

- Loosely coupled vs deeply integrated

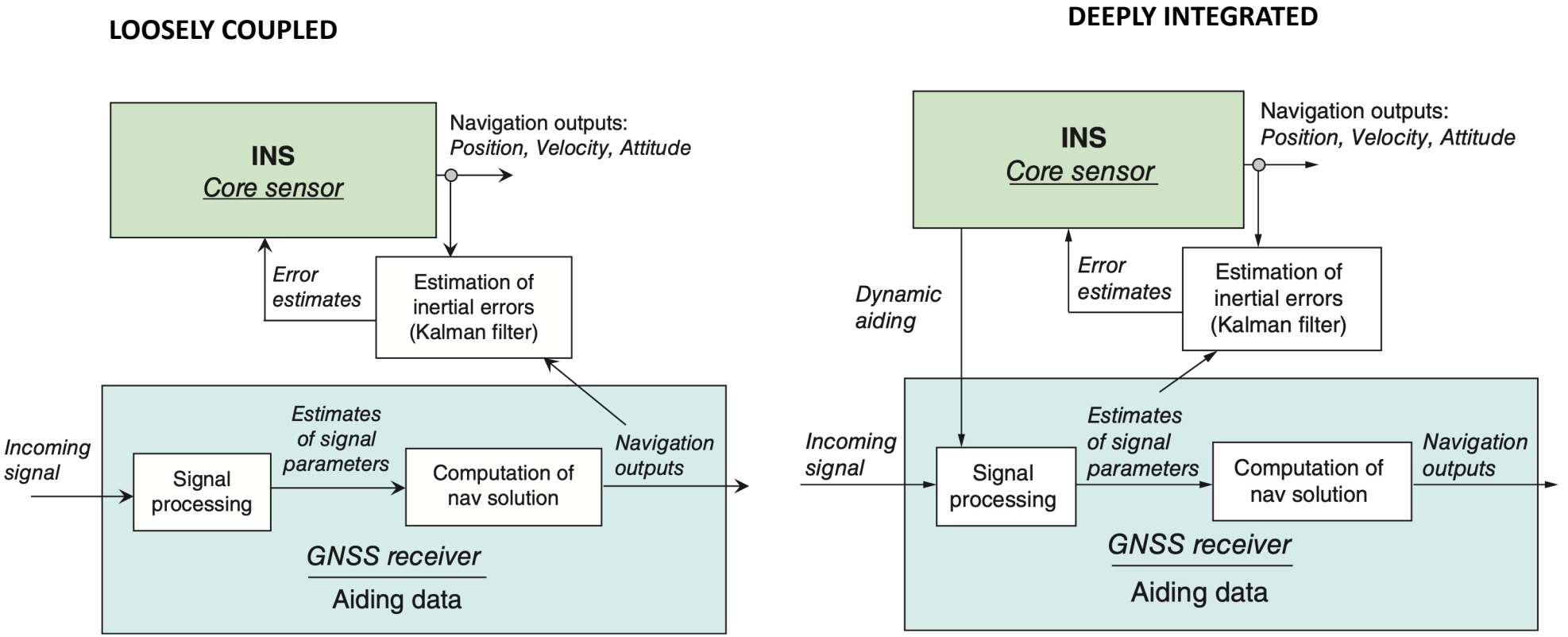


Figure 46.10 Loosely coupled GNSS/INS mechanization. GNSS navigation solution (v navigation outputs.

Figure 46.18 Deeply integrated GNSS/INS implementation. Deep integration extends tight coupling by incorporating inertial aiding of GNSS signal processing. This enables integration at the signal processing level, thus maximizing the benefits of sensor fusion.

Outlier detection in a coupled GNSS/INS system

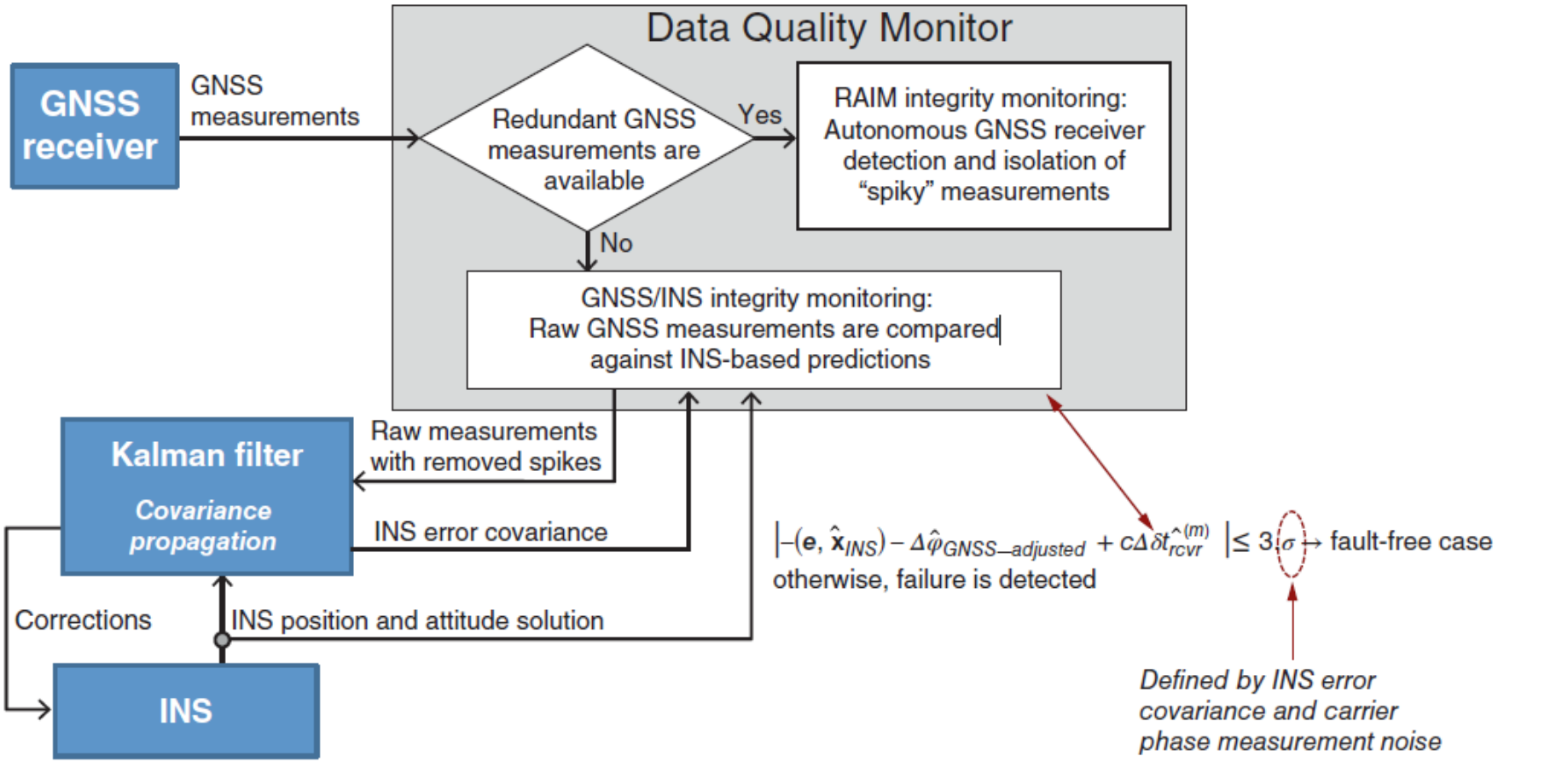


Figure 46.24 Measurement quality monitoring for detection and exclusion of outliers in GNSS measurements within the tightly coupled GNSS/INS architecture.

Urban Applications

Yang Wang

2024-11-20

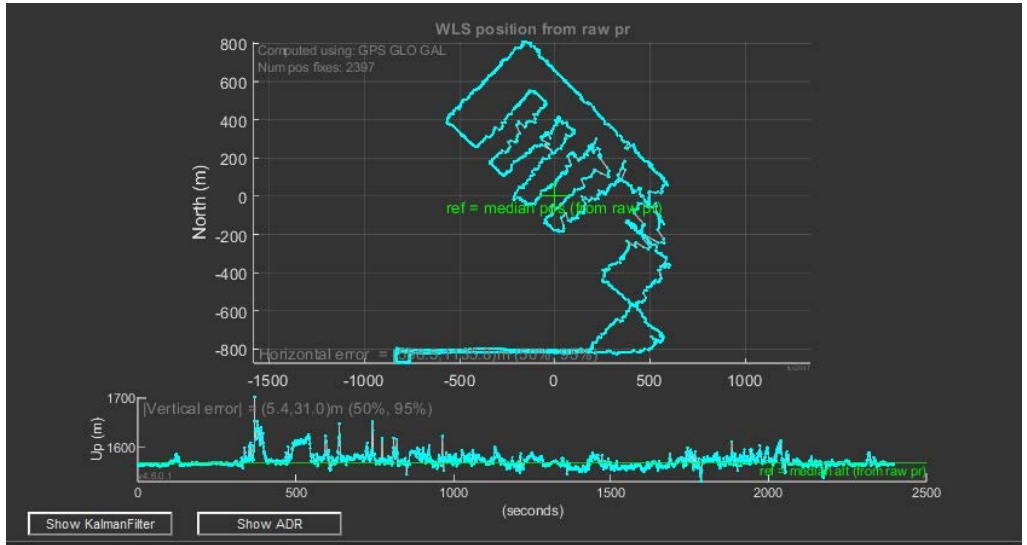


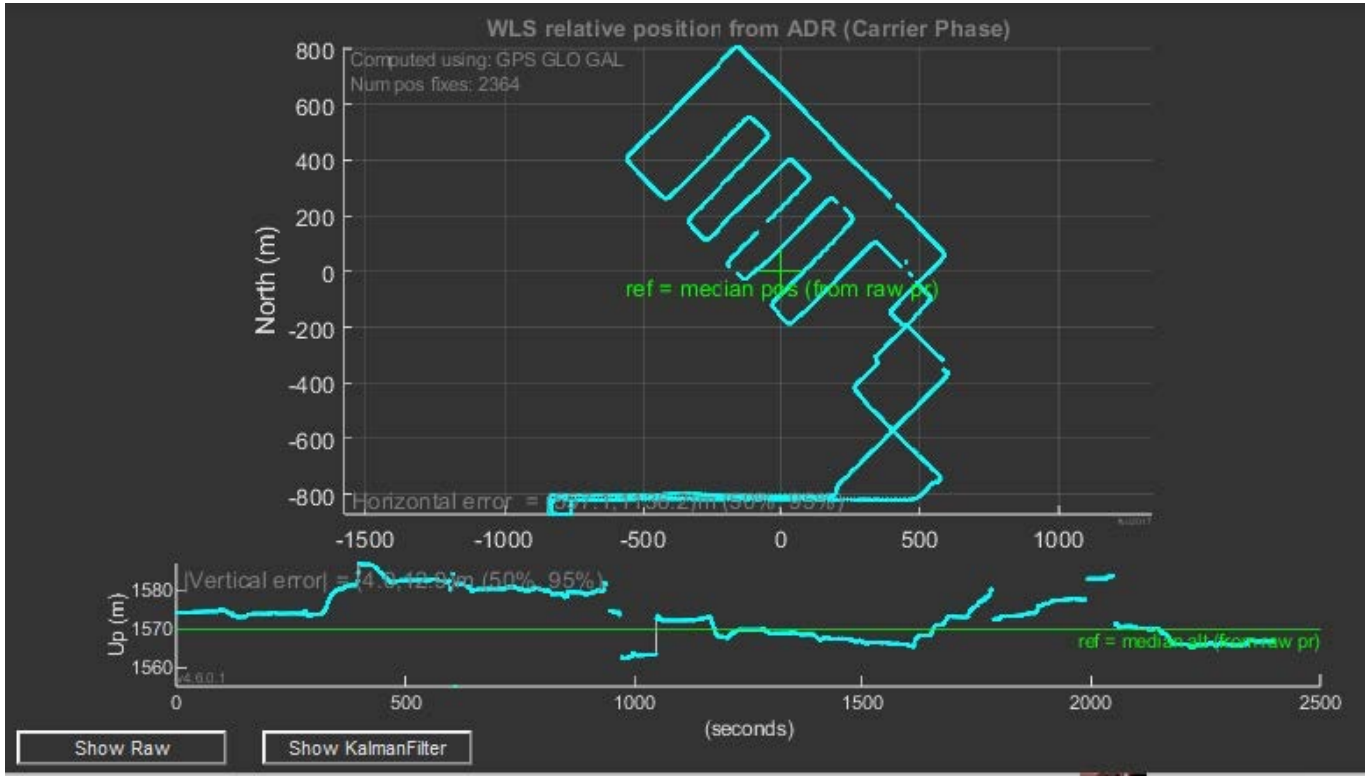
https://www.gpsworld.com/research-roundup-gnss-in-urban-canyons/

GNSS Urban Navigation Users

- Pedestrians – smartphone navigation, smartwatch, running apps, …
- Public transportation systems – real-time tracking
- Drivers – vehicle & smartphone navigation, ridesharing and delivery services, …
- Aviation (e.g., medical helicopters), drones, and autonomous vehicles, …
- Others – surveying, construction, smart city applications, etc.

GNSS Navigation with Kalman filtering





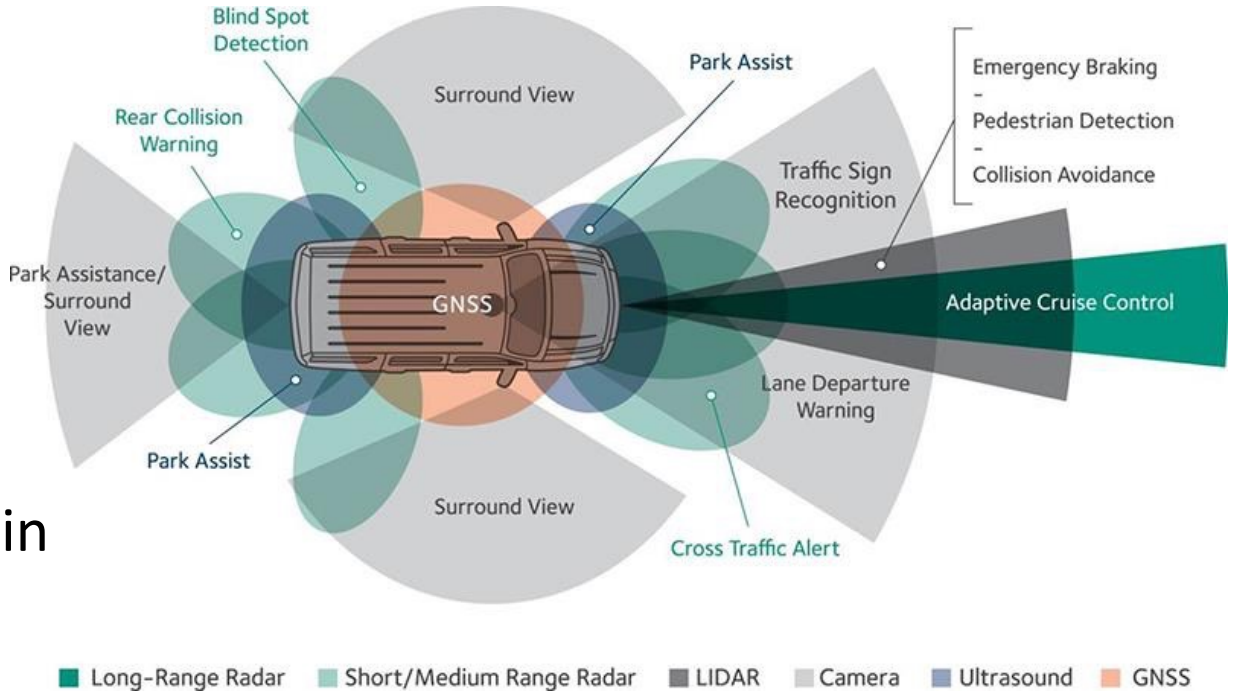
Credit: D.K. Lee

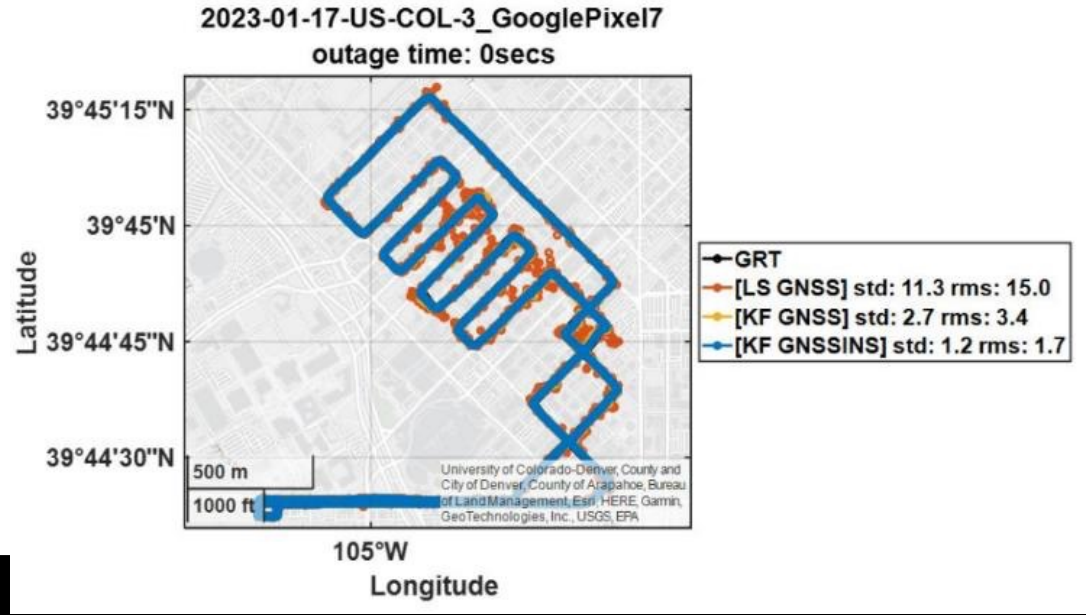
Sensor Fusion

Urban Navigation Sensor Availability

- Smart Vehicles (Safety is the concern)
  - GNSS Receiver with high-end antenna
  - IMU / Axle Counter / Camera / Lidar (Not in Tesla)
  - Tesla, Uber, Waymo, Cruise

- Smartphones (Price is the concern)
  - GNSS Receiver with cheap and small antenna
  - IMU / Barometer / Network Location / Camera
  - GNSS/INS sufficient for sub-meter level of





Credit: D.K. Lee

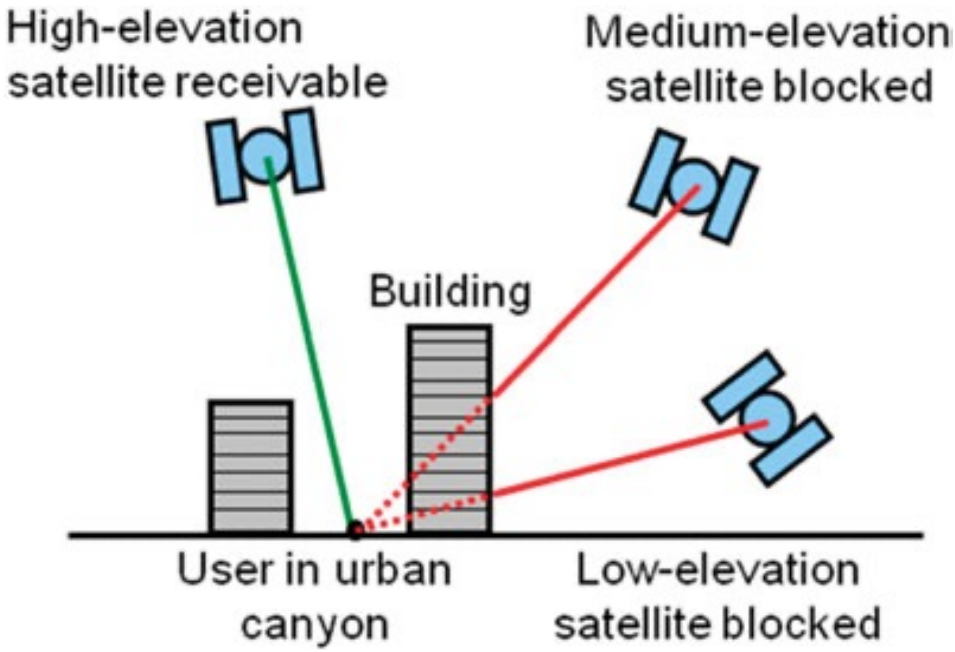
GNSS Urban Navigation Challenges

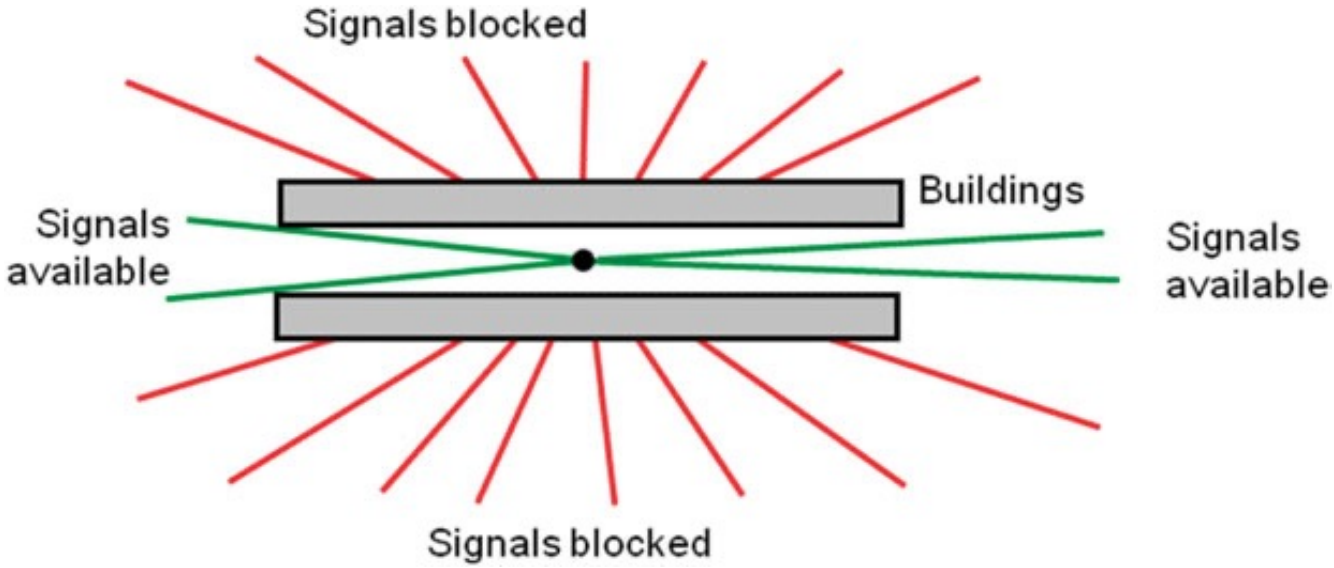
- Visibility – insufficient number of visible satellites and/or poor satellite geometry
- NLOS Signals and Multipath Effects
- Weak signal (e.g., blockage, diffraction, foliage attentuation)
- Radio Frequency Interference – intentional (jamming, spoofing) & unintentional (malfunctioning electronics, overlapping or adjacent band)

- Adversely impacting the GNSS accuracy, robustness, and integrity
- Advantages: additional RF sources for navigation, assisting information from networks through communication links

Shadow Matching

- A technique that finds most likely user location using the GPS satellite visibility predictions from the 3D model compared to the observed SNR values





P. Groves, 2011 (https://doi.org/10.1017/S0373463311000087)

Shadow Matching Scoring Scheme

- Shadow Matching demonstrates meter-level accuracies in the across-street direction in dense urban areas
  1. Select a search radius around the a priori position solution.
  2. Select a set of candidate locations
  3. At each candidate location, use the building model to predict the visibility of each satellite
  4. Scoring scheme compares the visibility predictions to the signal strength values.
  5. Each candidate location receives a score that represents the likelihood the GPS user is located at that particular position.

$$f_{pos}(p) = \sum_{s=1}^{m} f_{sat}(s, p, SS)$$

position candidates

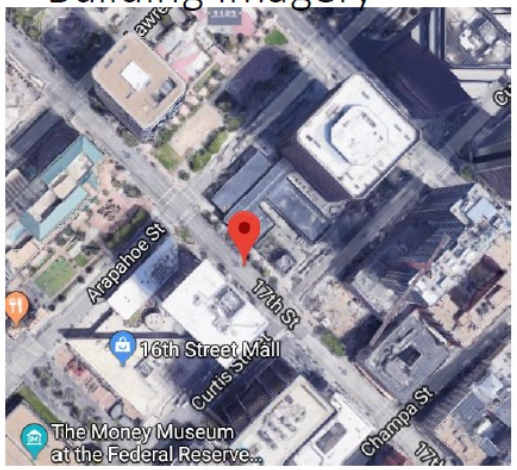
satellite

scoring scheme

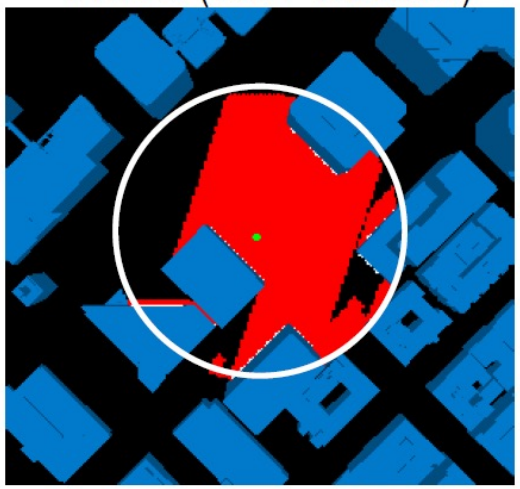
P. Groves, 2011 (https://doi.org/10.1017/S0373463311000087)

Shadow Matching Scoring Scheme - Binary

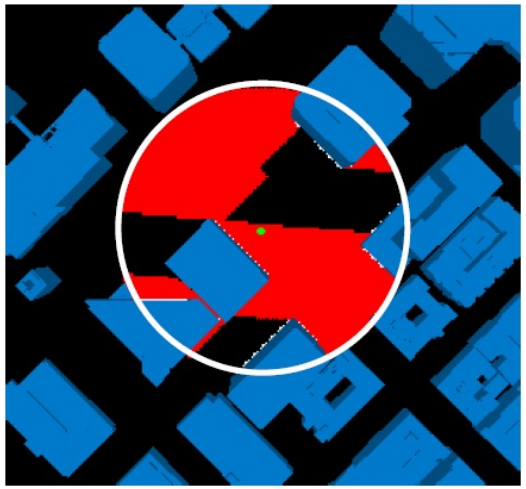
Building Imagery



PRN 3 (NOT Tracked)

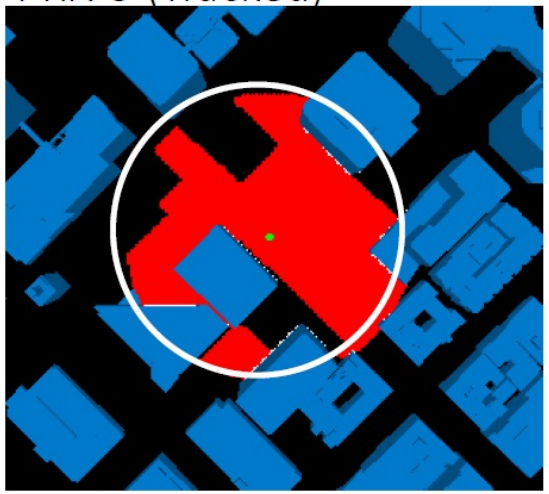


PRN 7 (NOT Tracked)

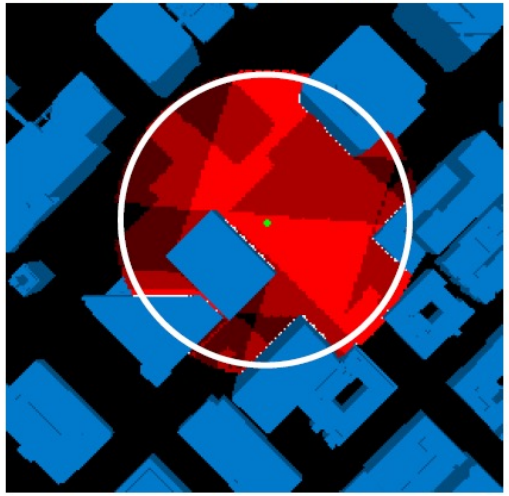


$$f_{pos}(p) = \sum_{s=1}^{m} f_{sat}(s, p, SS)$$

PRN 9 (Tracked)



PRN 3, PRN 7, PRN 9

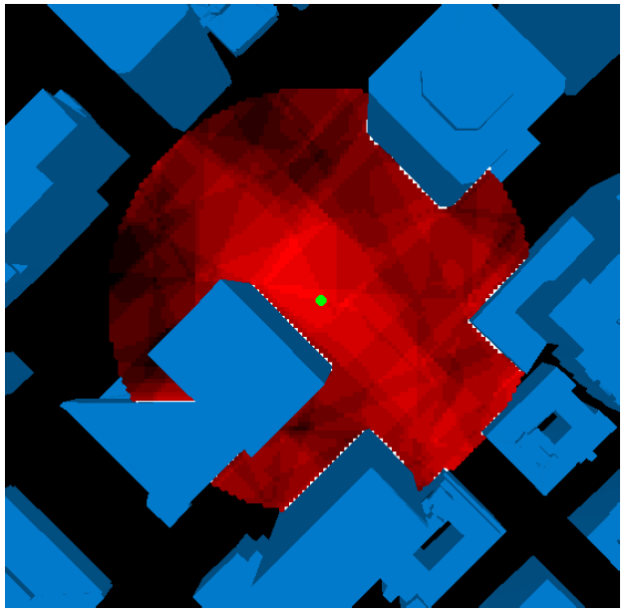




Low score

High score

Full constellation



Credit: K. Strandjord

Shadow Matching Scoring Scheme - Binary

- **BINARY**

Observation	Prediction: Invisible	Prediction: Visible
Not tracked	1	0
Tracked	0	1

(Groves, 2021)

- **TERNARY**

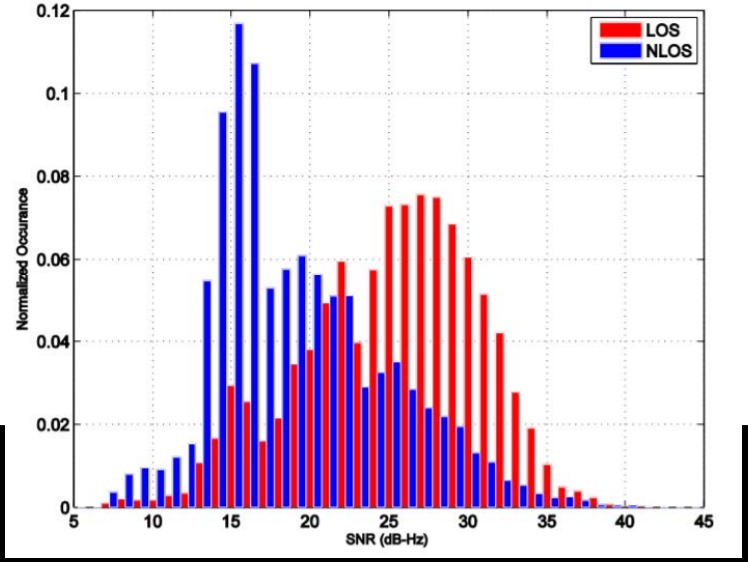
Observation	Prediction: Invisible	Prediction: Diffracted	Prediction: Visible
Not tracked	1	1	0
Weak signal	0.5	2	0.5
Strong signal	0	1	1

(Wang L., et al., 2012)

- **BAYESIAN**

$$p(LOS|SNR = s) = \frac{l_s}{l_s + n_s}$$

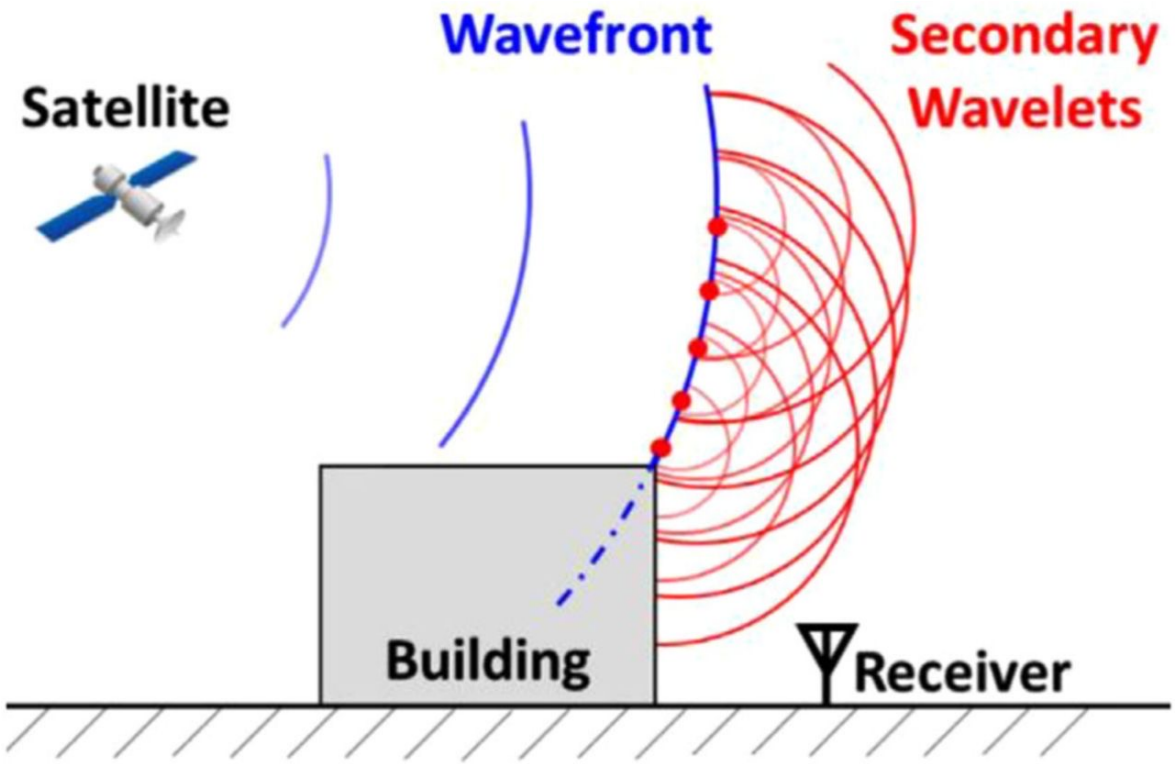
$$P_m = 1 - p(LOS|SNR = s) - b + 2b\,p(LOS|SNR = s)$$



(Wang L., et al., 2015)

Signal diffraction as described by the Huygens–Fresnel principle

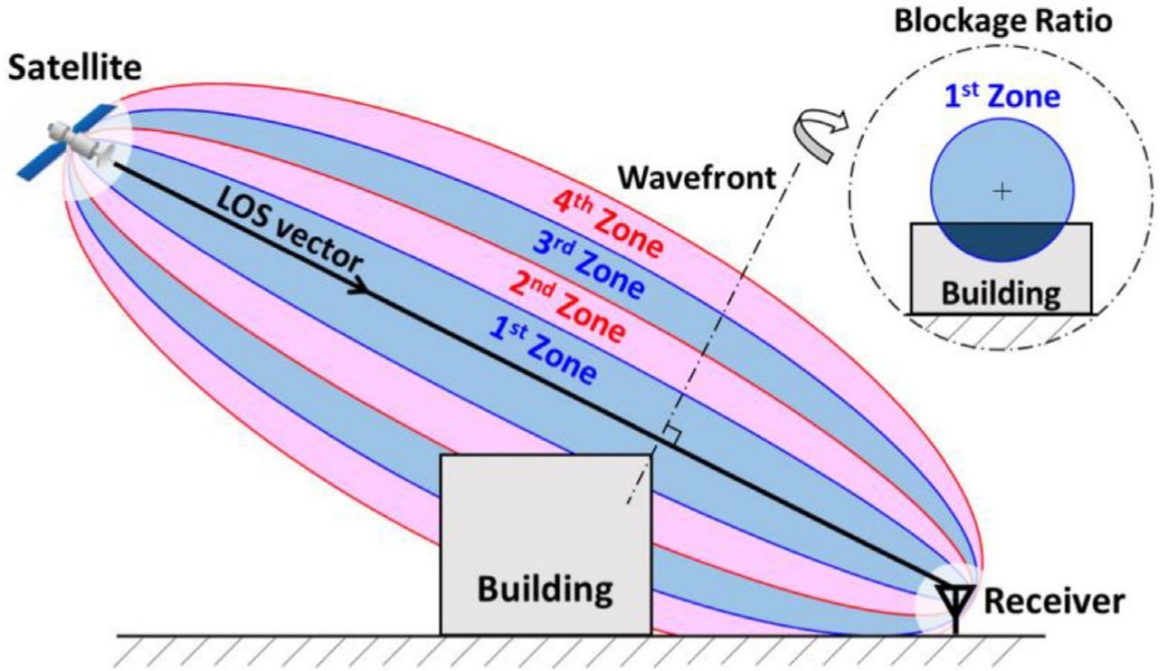
- Every point on a wavefront is itself the source of spherical wavelets, and the secondary wavelets emanating from different points mutually interfere.

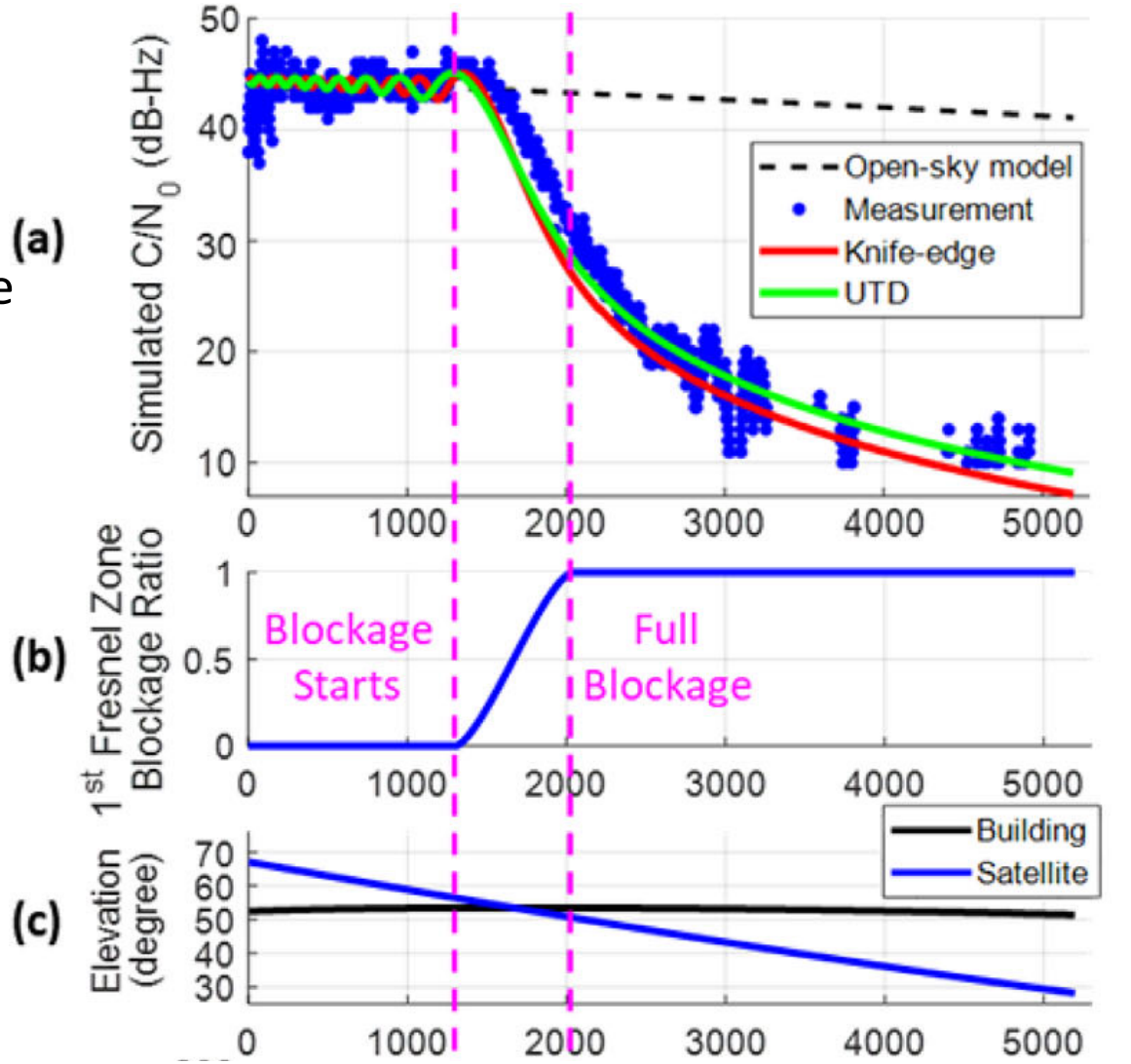


Zhang & Hsu, 2021 (DOI: 10.1002/navi.417)

Signal diffraction as described by the Huygens–Fresnel principle

- Fresnel Zones are defined as 3D elliptical regions between the transmit antenna and the receive antenna.
- The rule of thumb is that the primary Fresnel zone would ideally be 80% clear of obstacles, but must be at least 60% clear.

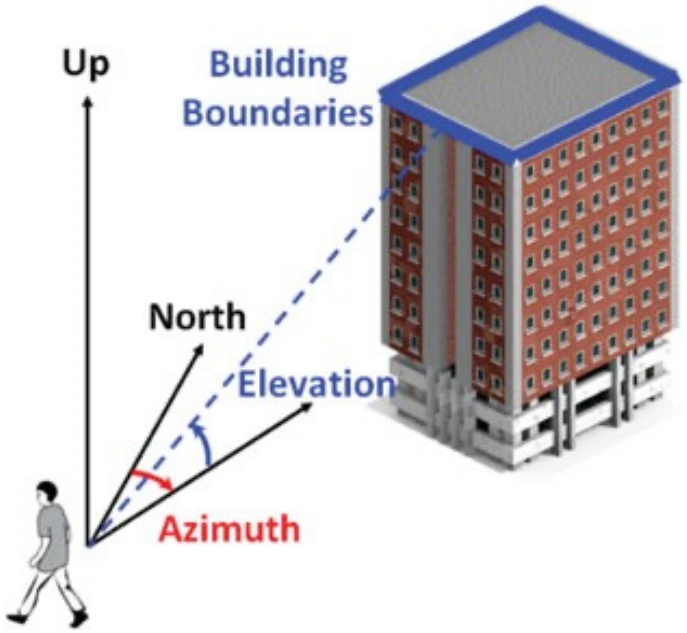


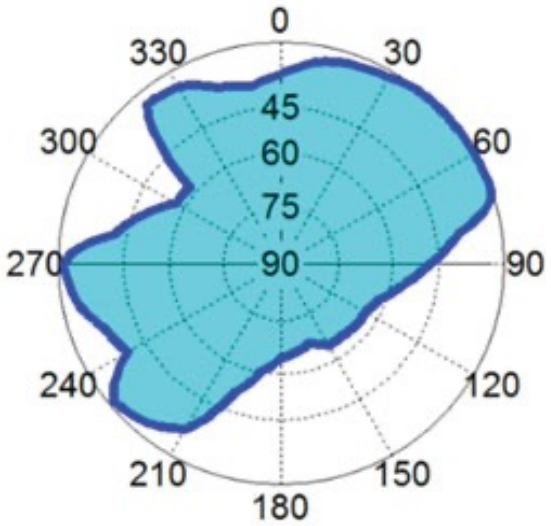


Zhang & Hsu, 2021 (DOI: 10.1002/navi.417)

3D-mapping-aided GNSS ranging with ray tracing

- Signal Weighting





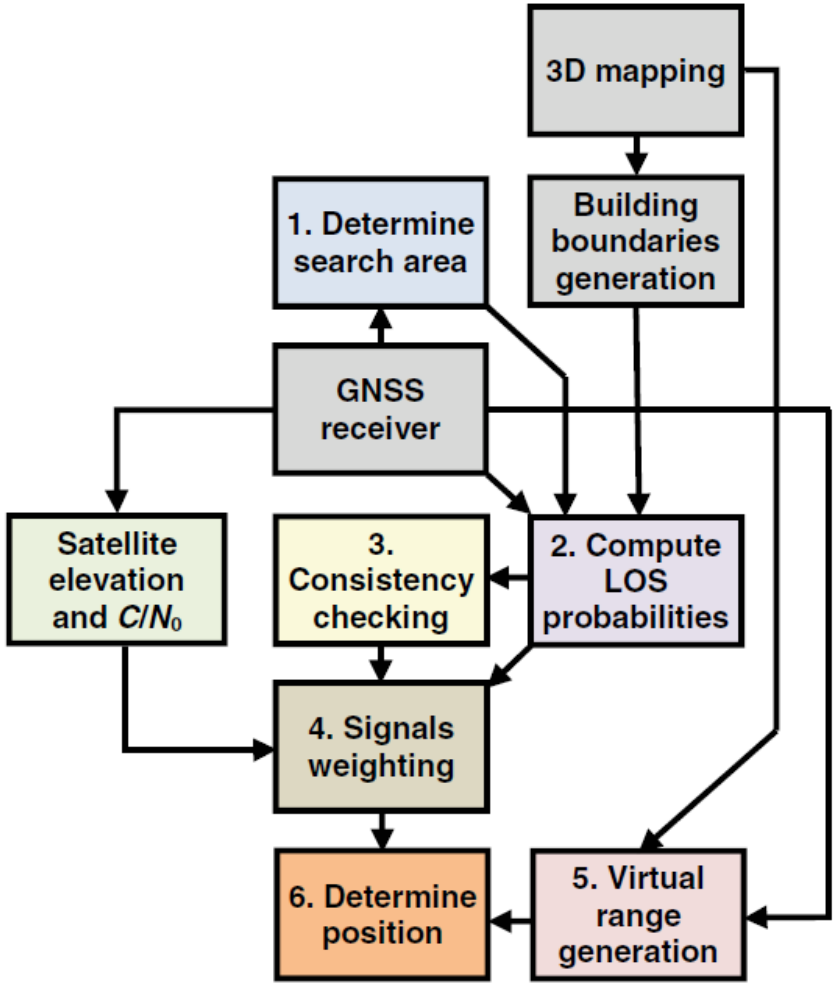


Figure 1. 3D-mapping-aided GNSS ranging algorithm block diagram (Adapted from Adjrad and Groves (2017)).

Adjrad & Groves, 2018 (https://doi.org/10.1017/S0373463317000509)

3D-mapping-aided GNSS ranging with ray tracing

- "End Game for Urban GNSS: Google's Use of 3D Building Models" – F. van Diggelen

Google has deployed 3DMA-GNSS in Android phones for almost 4,000 cities …

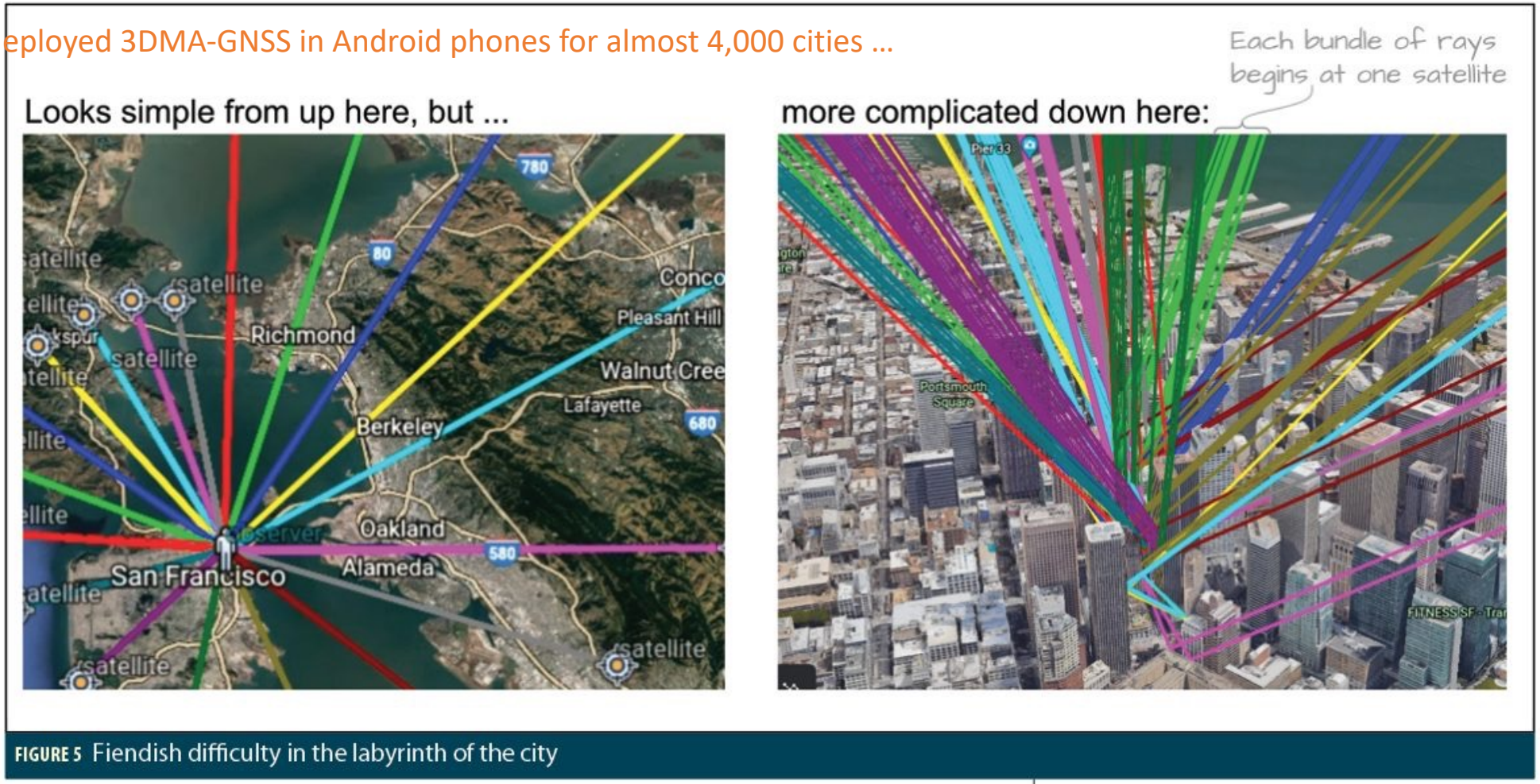


FIGURE 5 Fiendish difficulty in the labyrinth of the city

3D-mapping-aided GNSS ranging with ray tracing

- "End Game for Urban GNSS: Google's Use of 3D Building Models" – F. van Diggelen

We build a neural network (NN) connecting all our inputs (residuals) to the desired output (position). Each neuron is a linear combination of its inputs, (e.g. subtraction of residuals from different satellites) followed by an activation function (a non-linear operation, that can approximate any function, e.g. absolute value).
… ground truth in millions of training examples.

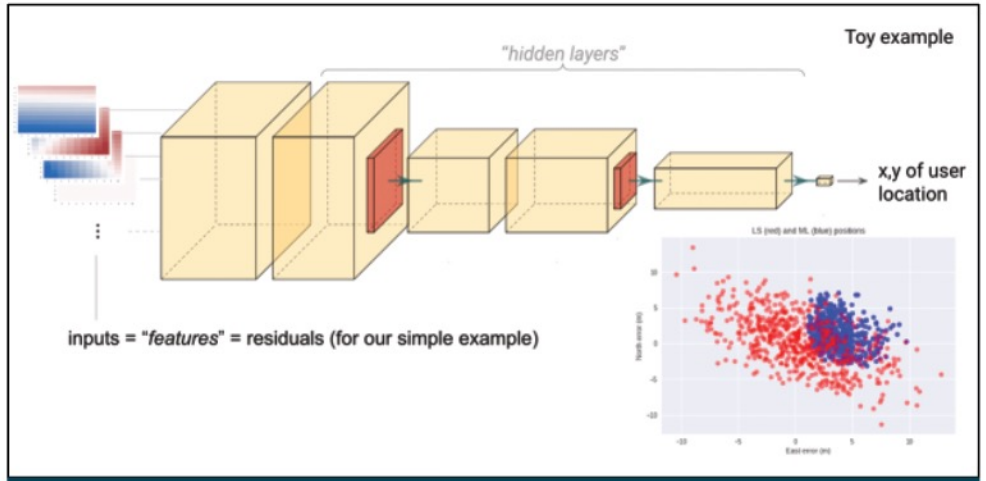


FIGURE 8 Neural network for machine learning

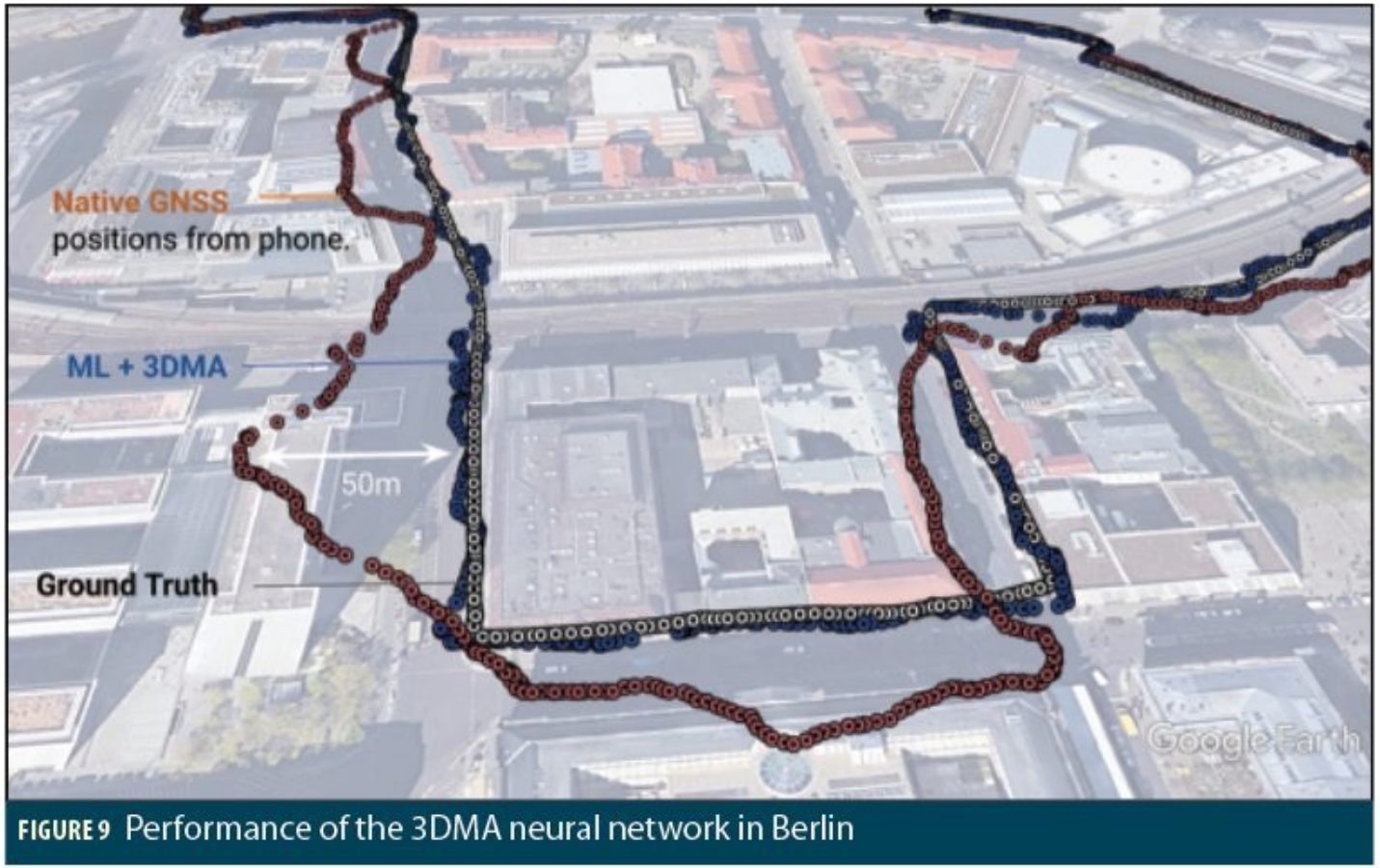
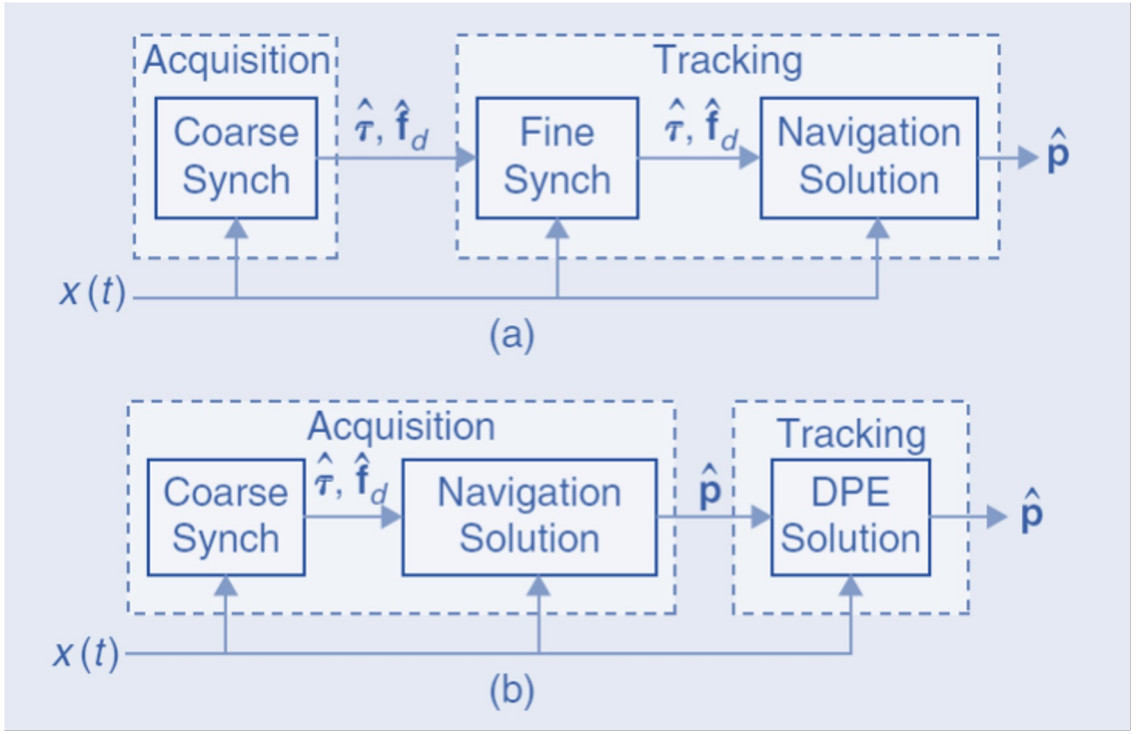


FIGURE 9 Performance of the 3DMA neural network in Berlin

Direct Position Estimation (DPE) Method

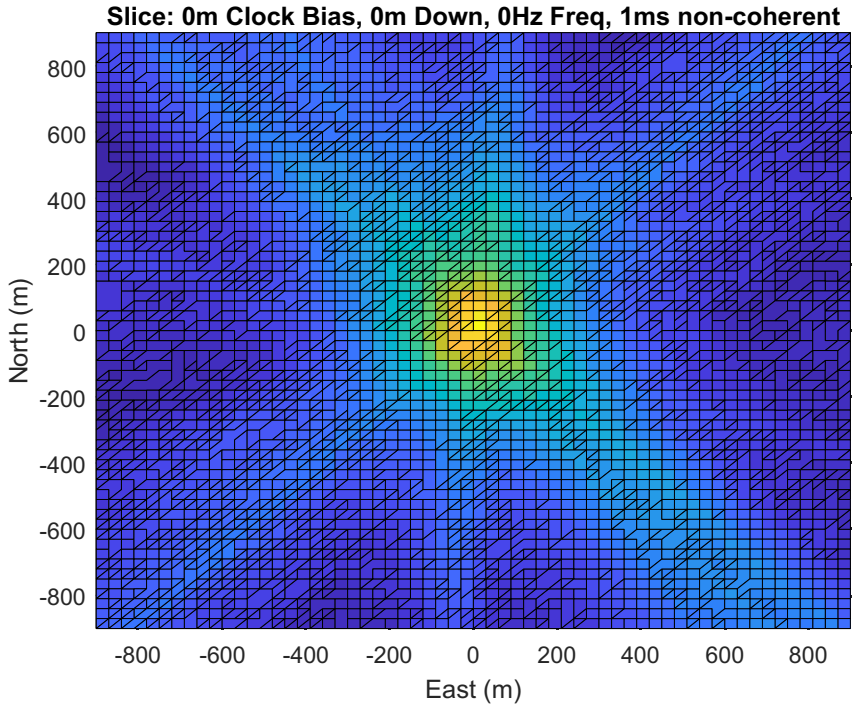
- Combines the acquisition and positioning processes



Conventional Receiver

DPE Receiver

DPE of GNSS Receivers (2017). Closas & Gusi-Amigo



Kirsten Strandjord

Conventional Acquisition: 2D Correlation

$$x_B[n] = \frac{1}{2}\sum_{i=1}^{nSV} A_i C_i[n - n_{i0}] D_i[n] e^{j\omega_{di} n t_s + j\phi_i} + noise$$

When acquiring a particular satellite $i = j$
We construct this reference signal:

$$x_{Rj}[n] = C_j[n - \hat{n}_{j0}] e^{-j\hat{\omega}_{dj} n t_s}$$

Correlation between the baseband input signal and the reference is:

$$x_B[n] \otimes x_{Rj}[n] =$$

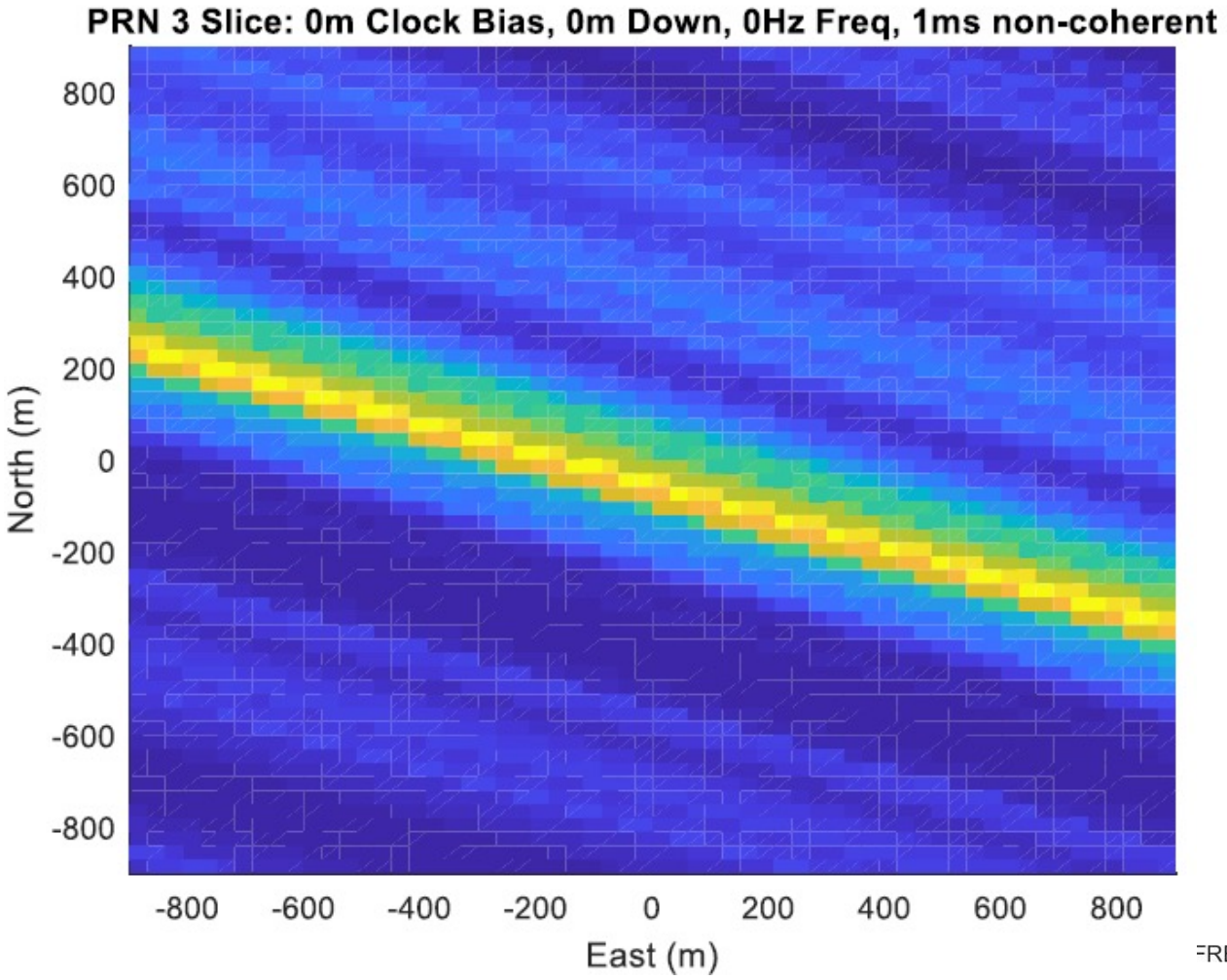
$$\frac{1}{2}\left(\sum_{i=1}^{nSV} A_i C_i[n - n_{i0}] D_i[n] e^{j\omega_{di} n t_s + j\varphi_i} + noise\right) \otimes \left(C_j[n - \hat{n}_{j0}] e^{-j\hat{\omega}_{dj} n t_s}\right)$$

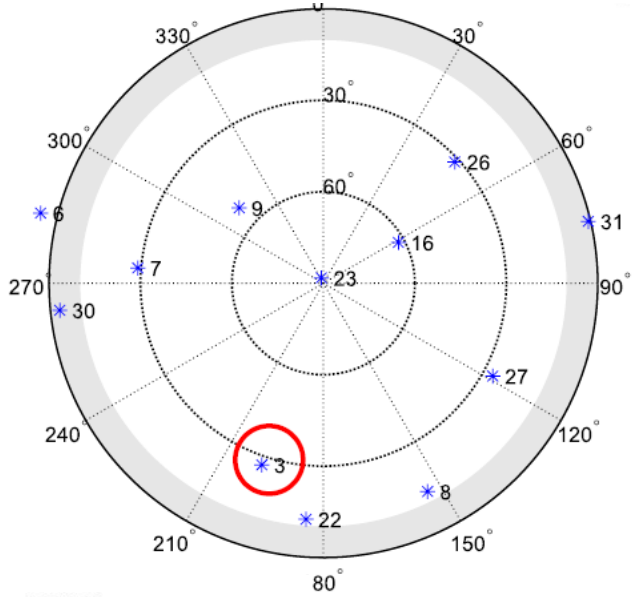
Acquisition with Direct Position Estimation (DPE)

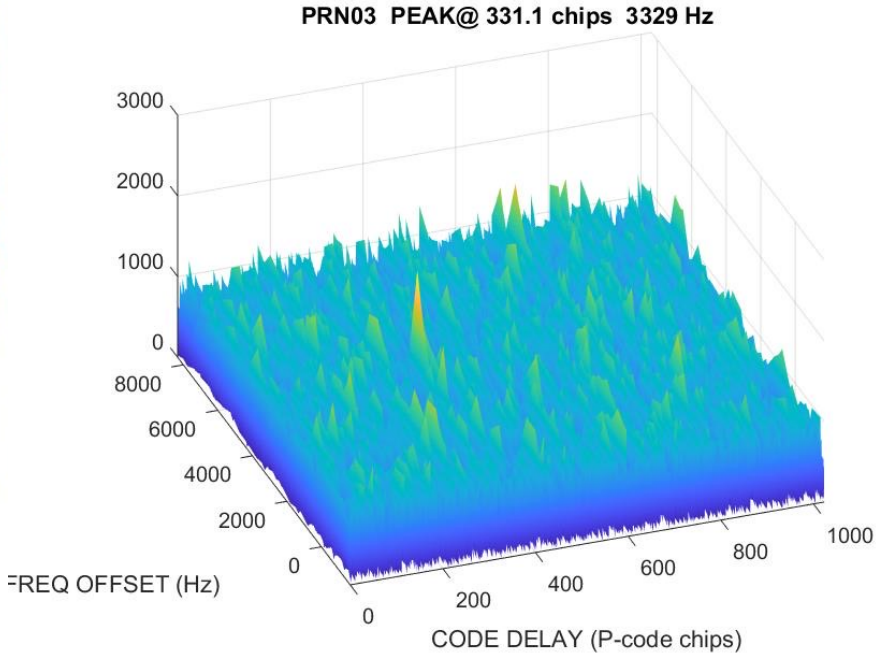
- Calculate the expected ranges and Doppler to each satellite at the a priori position and time.
- Create a correlogram of the received signal with replicas of each satellite over a grid of delay & Doppler consistent with the position search area
- At each candidate location and time:
    a. Compute the expected pseudoranges and apply the corrections.
    b. Pull the correlation values from the appropriate bin in the individual satellite correlograms
    c. Sum them to form the combined position domain correlogram.
- The peak of this position domain correlogram is the likely position of the user.

(Axelrad et al. 2011)

Acquisition with Direct Position Estimation (DPE)

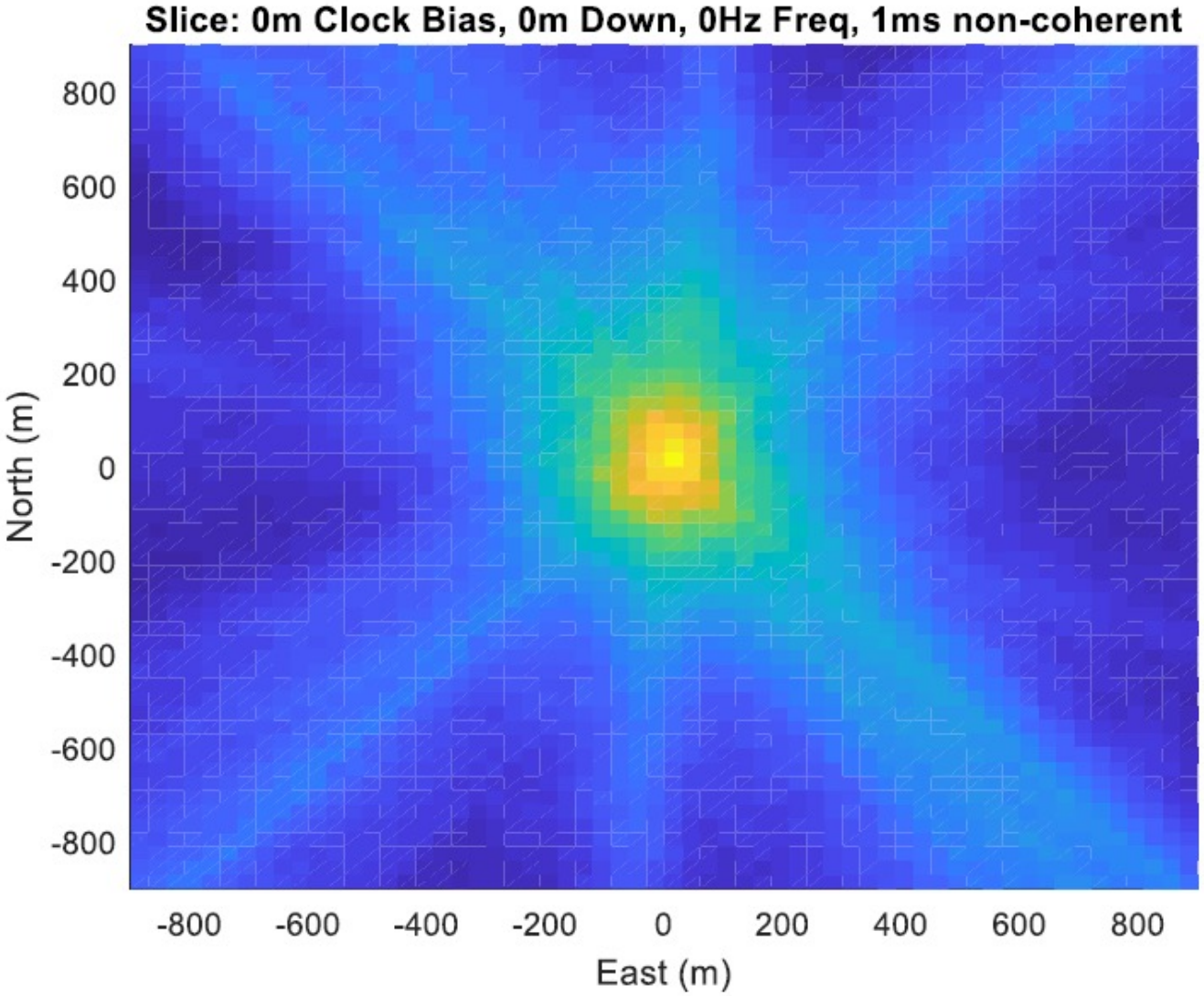


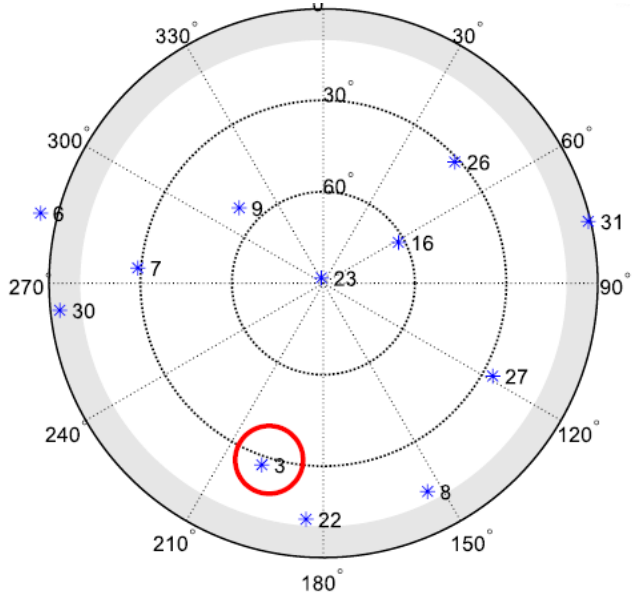




Acquisition with Direct Position Estimation (DPE)

Composite of Satellites





Navigation with Cellular Signals of Opportunity

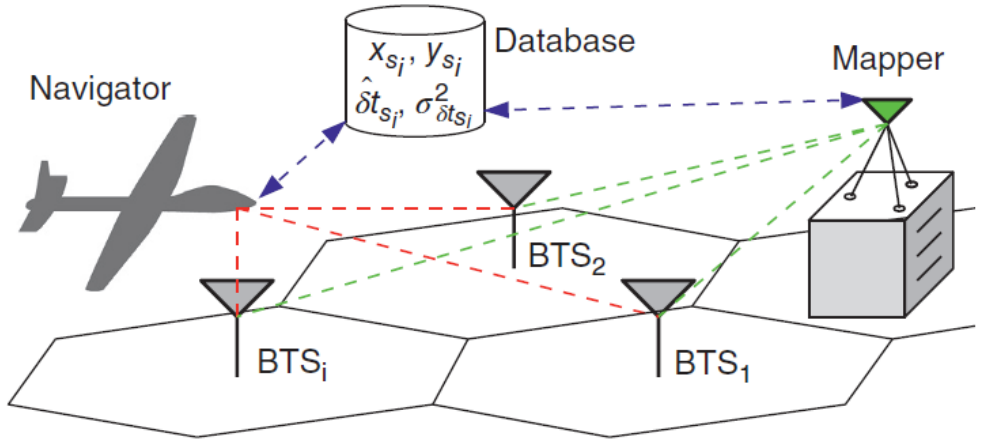
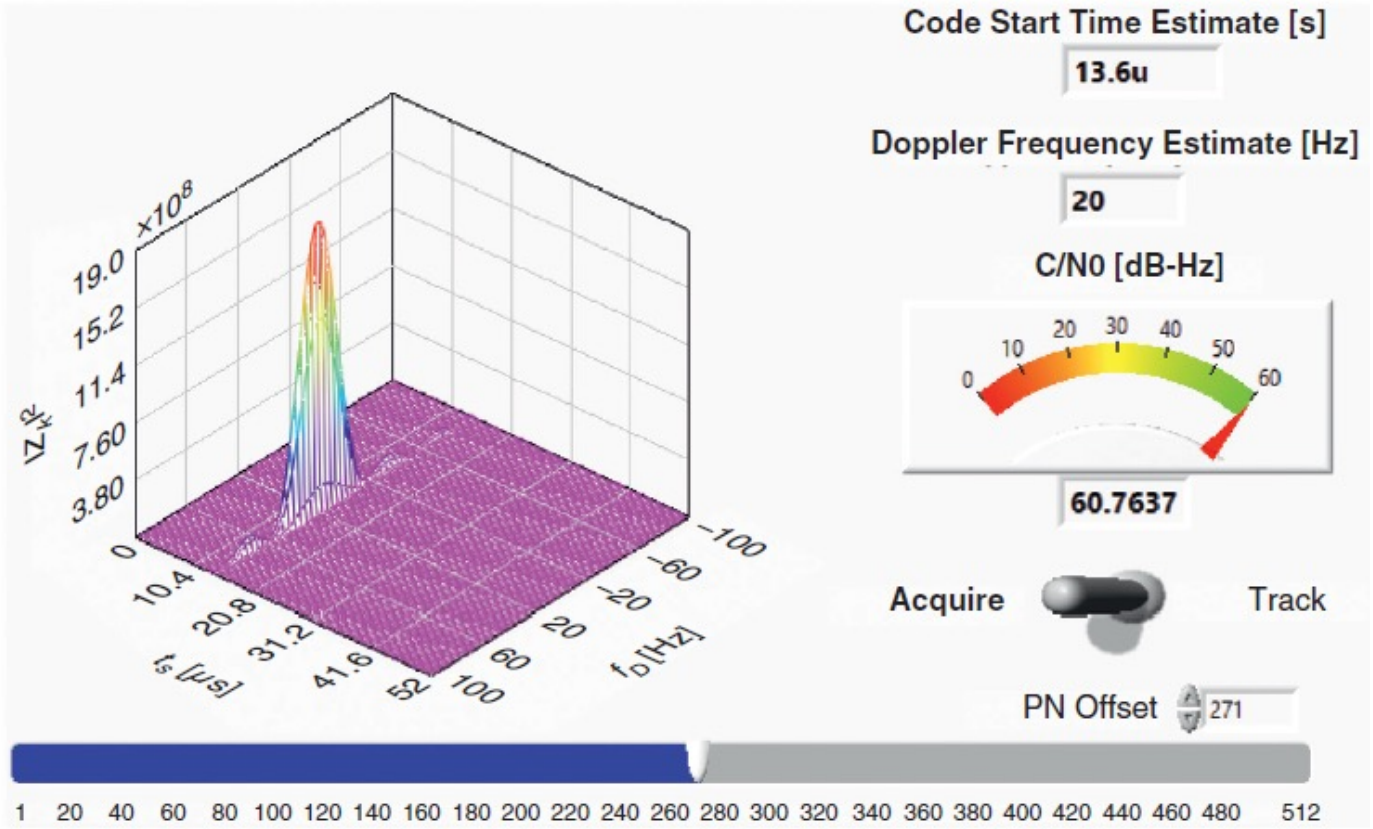


Figure 38.3 Mapper and navigator in a cellular environment (Khalife et al. [18]; Khalife and Kassas [25]). *Source:* Reproduced with permission of IEEE, ION.

An example of CDMA signal acquisition



Z. Kassas, Chp 38. 21st Century PNT

Navigation with Cellular Signals of Opportunity

- Some SoOp sources have similar signal and data structures with GNSS
- Follow similar navigation problem formulation
- Standalone & aiding modes

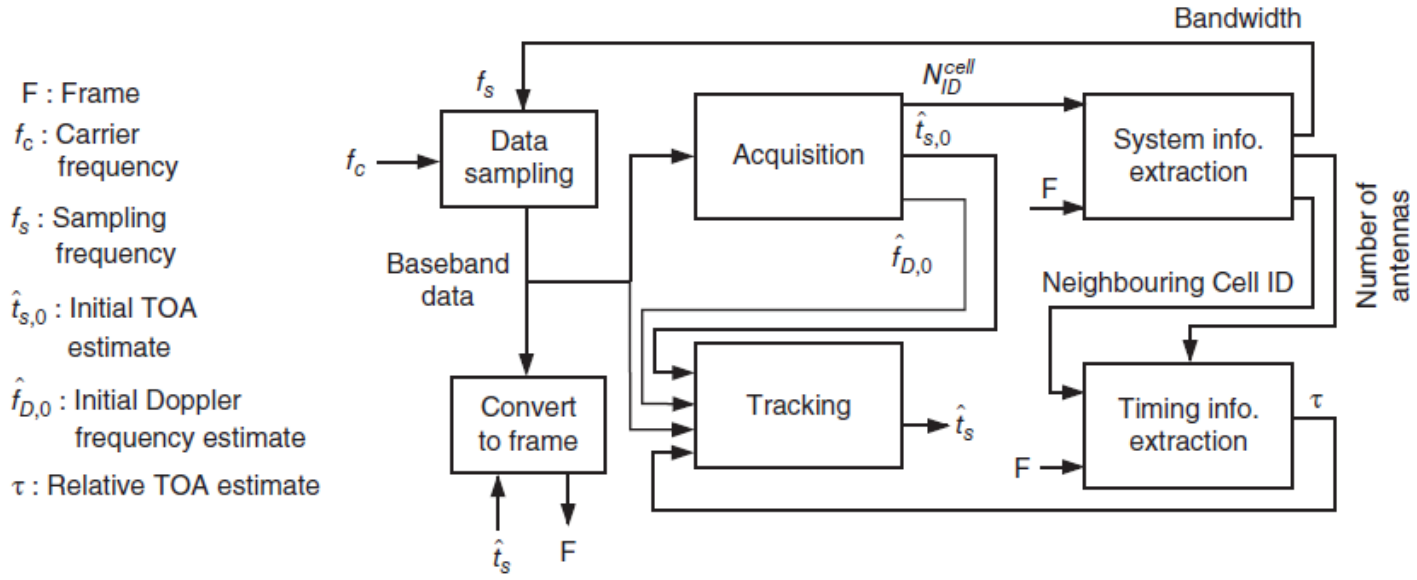


Figure 38.30 Block diagram of the LTE navigation receiver architecture (Shamaei et al. [65]). *Source:*

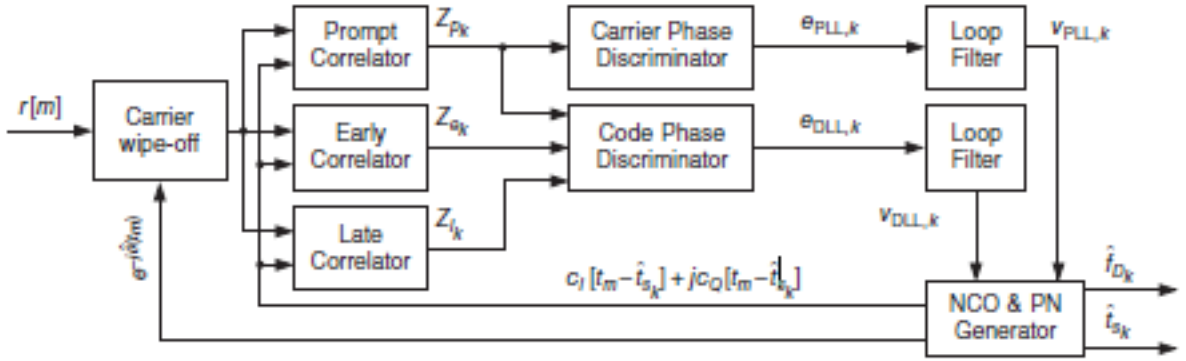


Figure 38.12 Tracking loops in a navigation cellular CDMA receiver. Thick lines represent complex quantities (Khalife et al. [18]).
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Next lecture: Midterm Review & HW Discussion